

A Virtual Reality Musical Instrument Integrated with a Remote Playing Robot System

Zhonghao Zhu*

Beijing Institution of Technology

Weizhi Nai†

Jilin University

Xin Wang‡

Jilin University

Yue Liu§

Beijing Institution of Technology

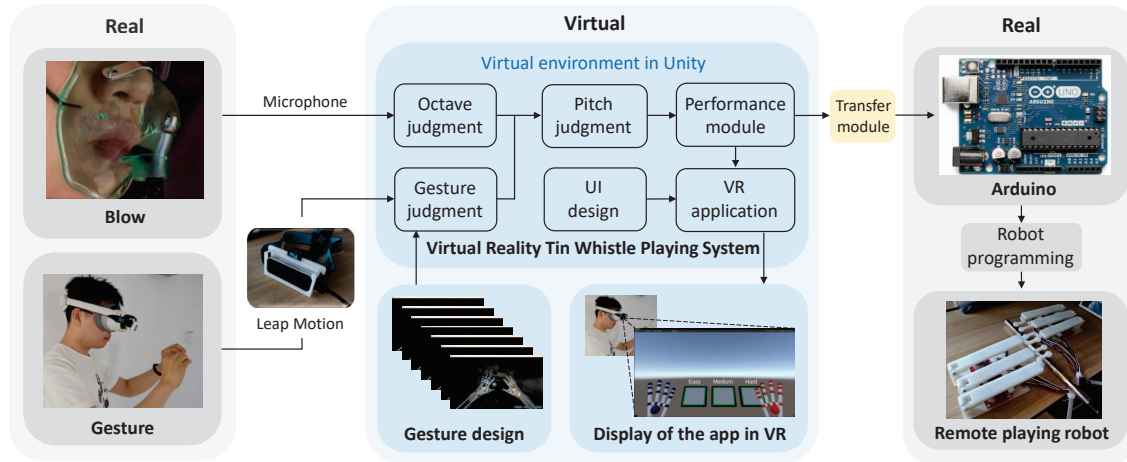


Figure 1: Research approach

ABSTRACT

Music education and performance are often constrained by geographical limitations. To evaluate the effectiveness of remote music performance, we designed a virtual reality musical instrument system with a remote playing robot. The system comprises a virtual Irish tin whistle playing system and a remote playing robot system. In the virtual playing system, sensors capture the performer's gestures, translate them into corresponding tin whistle playing commands and produce the music. The playing robot is constructed with mechanical hands, and a data transmission module is programmed to facilitate communication between the virtual playing system and the robot, enabling remote simulated performances.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Applied computing—Arts and humanities—Sound and music computing; Applied computing—Education—Distance learning;

1 INTRODUCTION

Music is a significant form of entertainment for humanity. The COVID-19 pandemic has led to a substantial decrease in the frequency and quality of collaboration among musicians, accompanied by an increased demand for online collaboration tools [1]. Compared to the pre-COVID-19 levels, the proportion of live music creation in social settings decreased by 79% during the period of social restrictions, while the proportion of online collaborative music creation

increased by 264% [2]. A virtual reality musical instrument (VRMI) emerges as a potential solution.

For the theoretical exploration of VRMIs, researchers have conducted on-site overviews from the performer's perspective, proposing nine design guidelines and detailed evaluation methods [4]. Additionally, comprehensive studies on music in Extended Reality (XR), including Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), have been undertaken [5]. These studies provide a feasible reference framework for the development of our virtual reality instrument.

Researchers have also made progress in the design of VRMI systems. AirFute, for instance, enables students to simulate playing musical notes on flute without the need for having a physical instrument [3]. However, AirFute lacks a physical playing system, which means it can only be used in virtual reality and is not suitable for remote live performances.

Therefore, to assess the effectiveness of remote music collaboration and teaching, we designed a virtual reality instrument system with remote playing functionality.

2 SYSTEM COMPONENTS

The VRMI and remote playing robot system have planned a pipeline of *real* → *virtual* → *real*, as depicted in Figure 1. The system comprises two subsystems:

- The process from the real to the virtual involves capturing necessary data such as gestures by sensors, followed by computation and assessment. Subsequently, the virtual environment facilitates the playing of the Irish tin whistle.
- The process from the virtual to the real involves transmitting the performance data to a microcontroller, which then controls the playing robot to produce real sounds from the Irish tin whistle.

*e-mail: 3120230531@bit.edu.cn

†e-mail: naiwz@jlu.edu.cn

‡e-mail: wxin@jlu.edu.cn

§e-mail: liuyue@bit.edu.cn ✉ Corresponding author

2.1 *Real*→*Virtual*: Virtual Reality Tin Whistle Playing System

2.1.1 Gesture design and microphone configuration

Figure 2 illustrates the playing techniques of the Irish tin whistle. Both fingering and blowing intensity work collectively to influence the pitch of the tin whistle.

In order to achieve better performance in the virtual reality whistle-playing system, a gesture recognition model is established to capture the whistle-playing actions. As shown in Figure 1, The gesture design module contains seven gestures, each of which corresponds to a consecutive note in a major scale. The Leap Motion API is utilized to retrieve vector data from the fingertips of the index, middle, and ring fingers of both hands.

A microphone is employed to assess the strength of breath based on the received volume. Among the techniques of the Irish tin whistle, breath intensity is categorized into three levels: no blowing, light blowing, and heavy blowing.

2.1.2 Virtual Tin Whistle Performance Module

The *Virtual* module in Figure 1 illustrates the workflow of the virtual playing system. In Unity3D, module for Leap Motion gesture recognition involves interfacing with the Leap Motion SDK to access parameters such as finger position, orientation, and speed. The fingertip distance is determined by subtracting the vector of the thumb's fingertip from the calculated vector, and the magnitude of this result is obtained. The fingertip distance is used to ascertain whether the sound hole is being covered.

Additionally, the *Octave judgment* module captures audio signals from the microphone device, measuring the user's blowing intensity based on the predetermined settings. The microphone is securely attached with a face mask to ensure accurate detection.

The *Pitch judgment* module assesses pitch based on the data from the former. The *Performance module* generates sound in the virtual environment according to the pitch, thereby achieving the playability of the virtual instrument.

2.1.3 Interaction Design and Display

The interaction design of the Virtual Reality Performance System is depicted in *Display of the app in VR* in Figure 1, featuring three modes: Easy, Medium, and Hard. The performance module associated with these modes caters to users with different playing levels. In the Easy mode, sound production and pitch modulation are controlled solely through gestures. The Medium mode incorporates breath analysis to determine pitch, while the Hard mode relies entirely on the user's breath for sound production and pitch modulation.

Compared to flat displays, the 3D display effect on VR headsets is more impressive. This research explores the possibilities of virtual performance within VR headsets, which enhances the user's sense of immersion.

2.2 *Virtual*→*Real*: Remote Playing Robot System

Remote playing extends the reach of music education beyond geographical constraints and also explores the potential of online live music performances. Musicians can perform in their own locations, when the remote playing robot precisely replicates their actions, producing sounds from the instrument as if played live by themselves.

2.2.1 Construction of a Robotic System

The robot system comprises six mechanical fingers (corresponding to the six finger holes of the Irish tin whistle), an air pump module, and a bottom plate for fixation and installation. The movable and immovable components of the mechanical fingers are connected using bearings and shafts, and they form an interference fit with 3D-printed parts to achieve precise motion control. Electromagnets are installed at the midpoint of the immovable components,

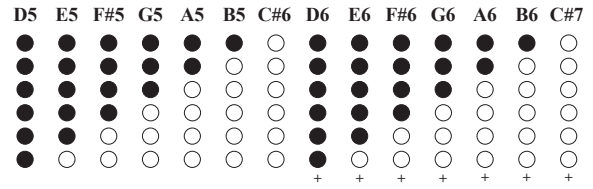


Figure 2: Fingering chart for a D key Irish tin whistle, where black dots represent covered finger holes, white dots represent open finger holes, and "+" indicates heavy blowing.

and magnetic strips are placed beneath the movable components. When the electromagnets are powered, the movable components descend, simulating the act of covering the finger holes. When the electromagnets are turned off, springs reset the movable components.

2.2.2 Data transfer module and robot programming

Using the Ardity plugin, we designed a data transmission module to transfer performance data to an Arduino microcontroller. Upon receiving the performance data, the robot, guided by a pre-defined program, determines and executes playing gestures to produce real sounds from the Irish tin whistle.

3 CONCLUSION

This study has implemented a system that combines a virtual reality instrument with a remote playing robot. Users can learn the tin whistle in a virtual world through the virtual playing system and engage in remote performances using the remote playing robot. This research opens up new possibilities for applications such as remote music education, remote music collaboration, and remote music concerts.

ACKNOWLEDGMENTS

The work was supported by science and technology innovation 2030—major project of "New Generation Artificial Intelligence" (Grant No. 2022ZD0115901) and the National Natural Science Foundation of China (No.61960206007) and the 111 Project (B18005)

REFERENCES

- [1] N. R. Fram, V. Goudarzi, H. Terasawa, and J. Berger. Collaborating in isolation: Assessing the effects of the covid-19 pandemic on patterns of collaborative behavior among working musicians. *Frontiers in Psychology*, 12:674246, 2021. doi: 10.3389/fpsyg.2021.674246
- [2] K. E. Onderdijk, F. Acar, and E. Van Dyck. Impact of lockdown measures on joint music making: Playing online and physically together. *Frontiers in Psychology*, 12:642713, 2021. doi: 10.3389/fpsyg.2021.642713
- [3] K. Sanborn. Airflute: A virtual flute and tutoring system. In *Proceedings of the 2021 ACM Southeast Conference*, ACM SE '21, p. 246–248. Association for Computing Machinery, New York, NY, USA, 2021. doi: 10.1145/3409334.3452087
- [4] S. Serafin, C. Erku, J. Kojs, N. C. Nilsson, and R. Nordahl. Virtual Reality Musical Instruments: State of the Art, Design Principles, and Future Directions. *Computer Music Journal*, 40(3):22–40, 09 2016. doi: 10.1162/COMJ.a.00372
- [5] L. Turchet, R. Hamilton, and A. Çamci. Music in extended realities. *IEEE Access*, 9:15810–15832, 2021. doi: 10.1109/ACCESS.2021.3052931